

power supply design

the main idea

Stage 1 : Transformer : The AC source gives 150V AC. The transformer steps it down 10:1, so you get 15v out. AC means the voltage goes up and down like a wave (sine wave)

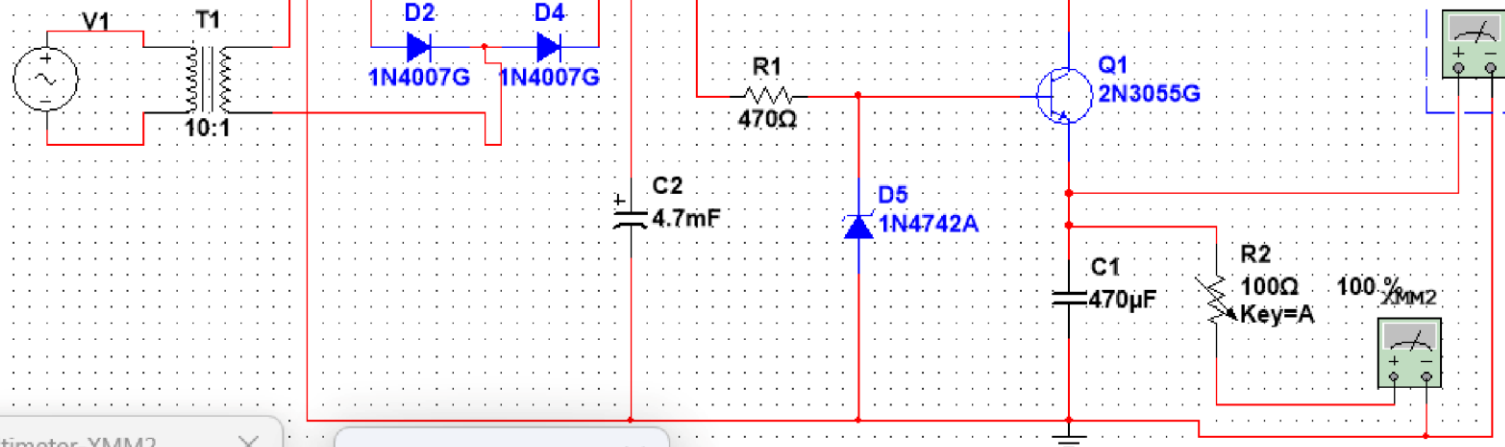
Stage 2 :Bridge Rectifier The 4 diodes flip the negative half of the sine wave to positive. Now instead of a wave going + and –, you get a wave that only goes + . This is called **pulsating DC**

Stage 3 : Filter Capacitor C2 The big capacitor smooths out the wave. It charges up when voltage is high, and releases energy when voltage is low. The result is an almost flat dc voltage

Stage 4 : Voltage Regulator (Zener + Transistor) We have a voltage come from transformer but we want to cut it at certian dc value This is where the Zener and transistor bring it down to a stable output.

C1: is the output filter capacitor placed directly across the load, after the transistor's emitter.

150Vrms
50Hz
0°



Multimeter-XMM2

113.498 mA

A

V

Ω

dB

~

-

Set...

Multimeter-XMM1

11.35 V

A

V

Ω

dB

~

-

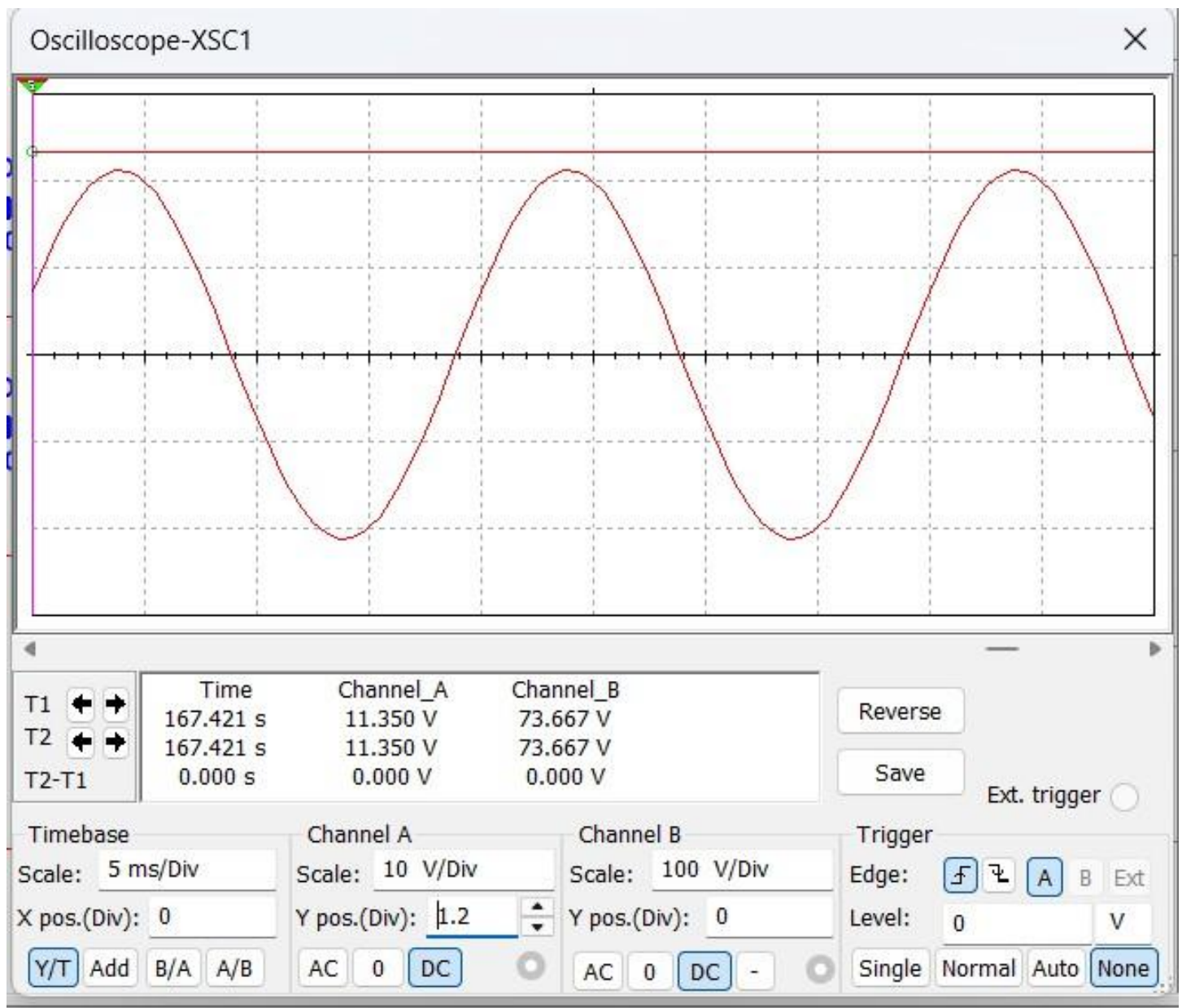
Set...

Choice of Component Values

Choice of C2 = 4.7 mF : C2 must be large enough to act as a charge reservoir it charges up to the peak voltage and slowly releases energy between rectifier pulses, preventing the voltage from dropping significantly between cycles , although C2 must not be excessively large because a very large capacitor increases the charge time constant ($\tau = RC$), meaning it takes longer to reach peak output voltage at startup.

Choice of R1 = 470 Ω : R1 must be small enough to allow sufficient current to flow through the Zener diode and into the base of the transistor. If R1 is too large, the Zener does not receive enough current to enter its breakdown region also R1 must be large enough to limit the current through the Zener diode and prevent it from exceeding its maximum power rating, zener diode wants at maximum 83 mA and by choosing these resistance value it will provide about 16.6 mA , so it will work safely.

Choice of C1 = 470 μ F: Fast output transient response, $\tau = 0.047s$, suppresses voltage spikes from sudden load changes

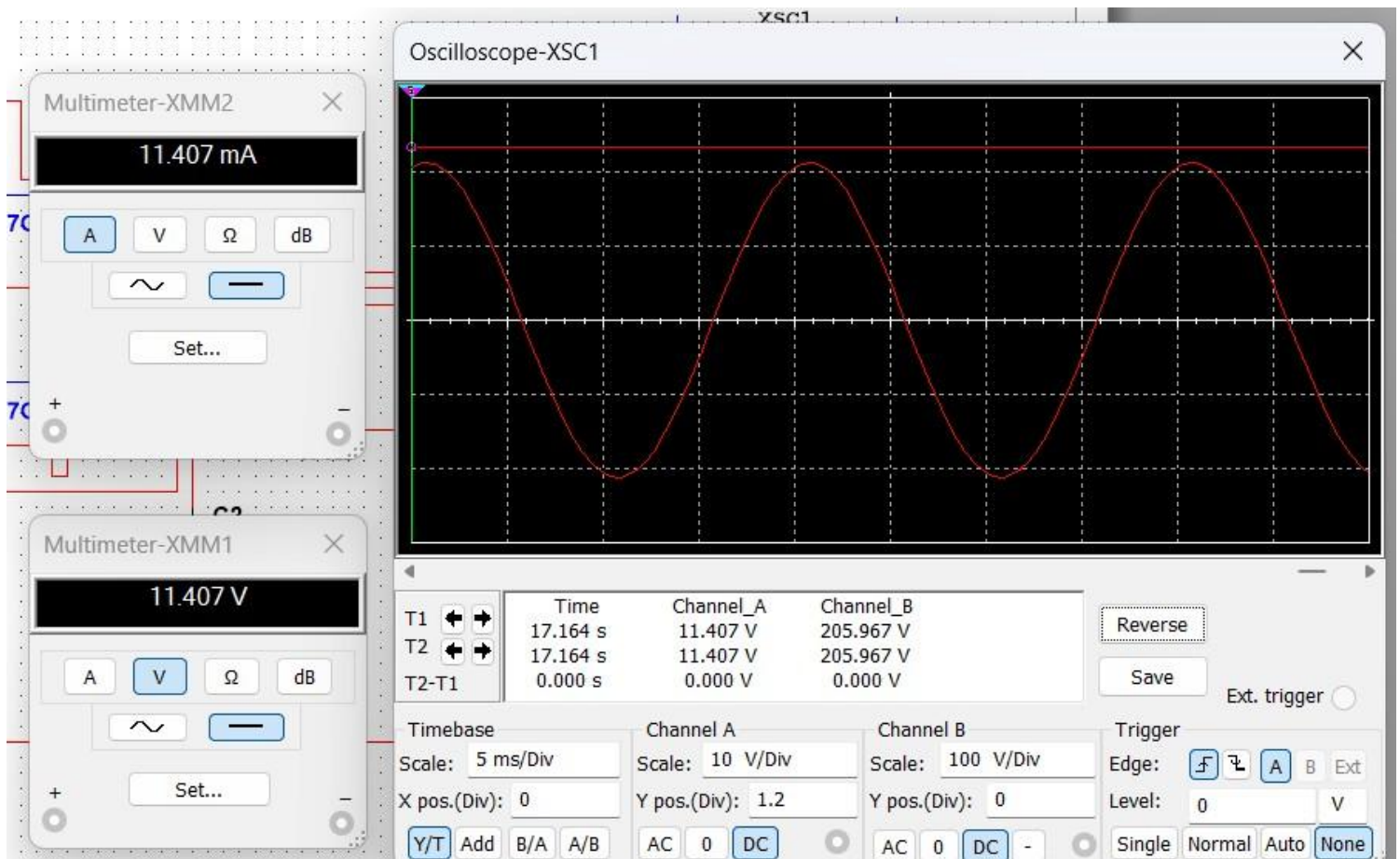
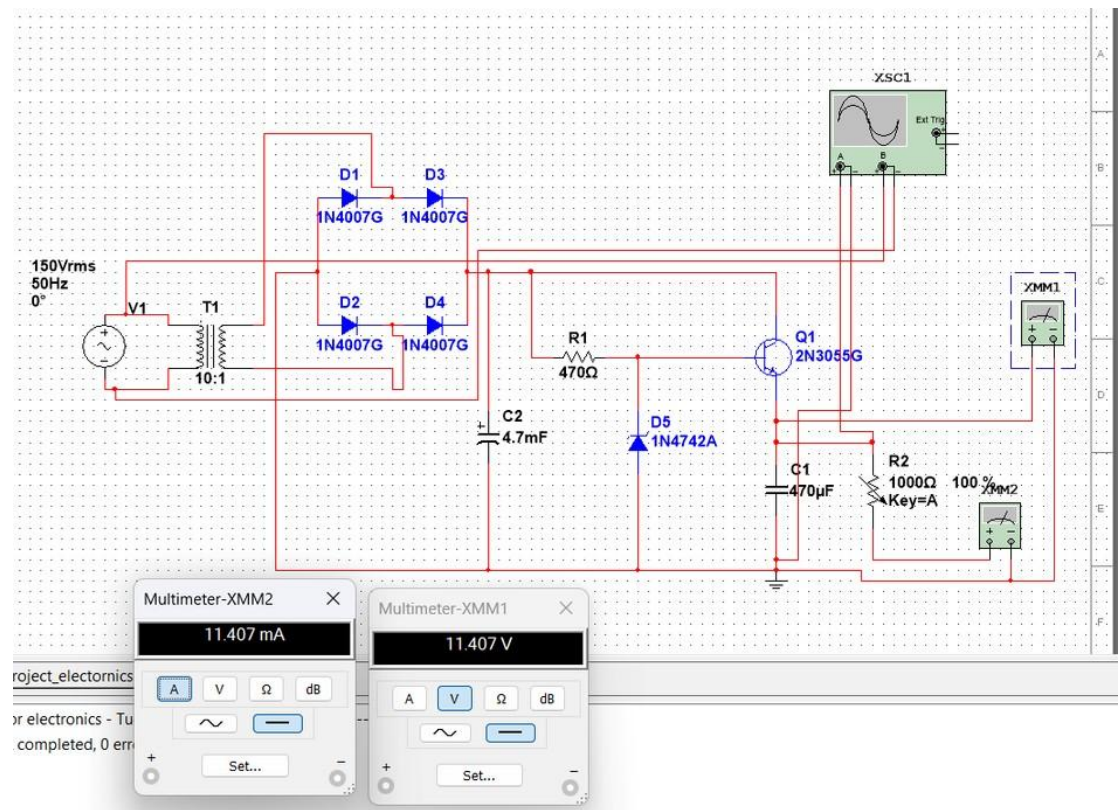


Why is the Output Not Exactly 12V?

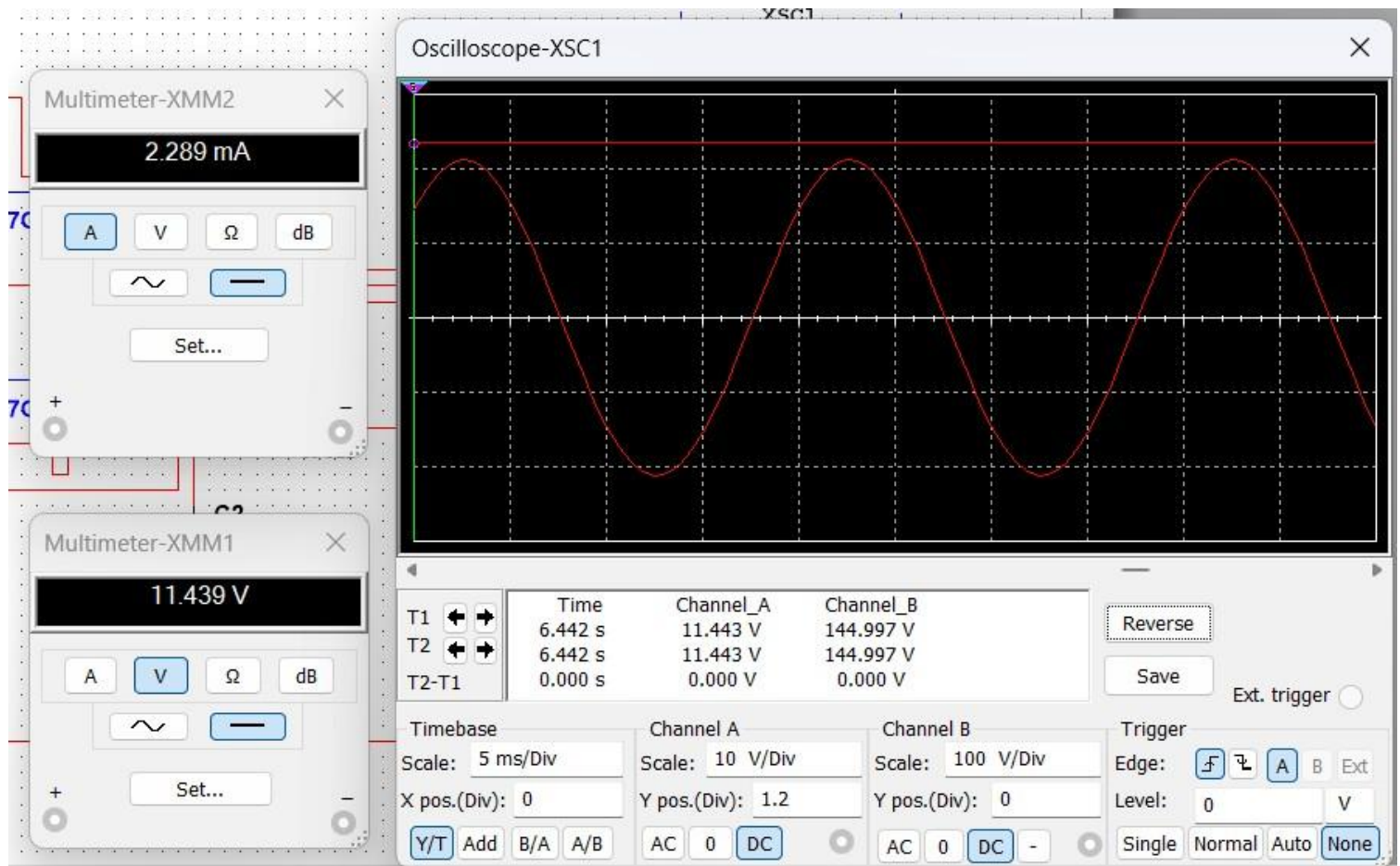
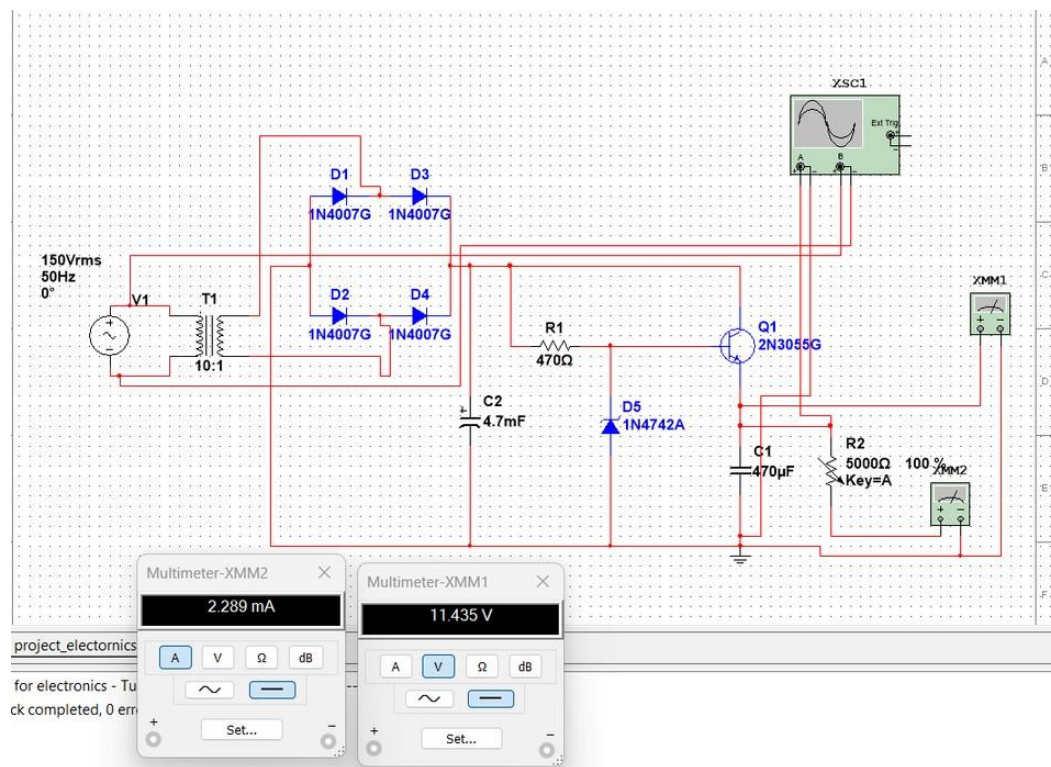
The Zener diode holds exactly 12V at its terminals. But the output doesn't come directly from the Zener it comes from the transistor's emitter, This difference is called the base-emitter voltage drop V_{BE} and it's always from 0.6 to 0.7 in silicon transistor

$$V_{out} = V_{Zener} - V_{BE} = 12 - 0.65 = 11.35V$$

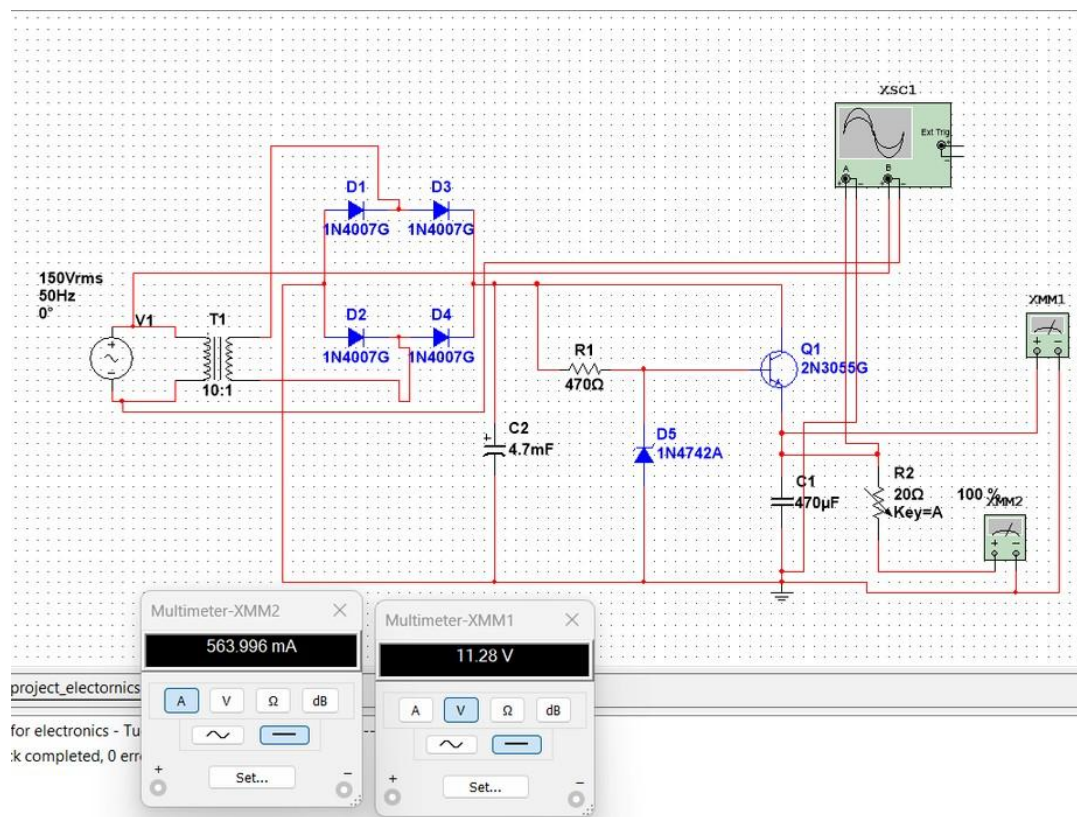
putting the variable resistance R2=1000



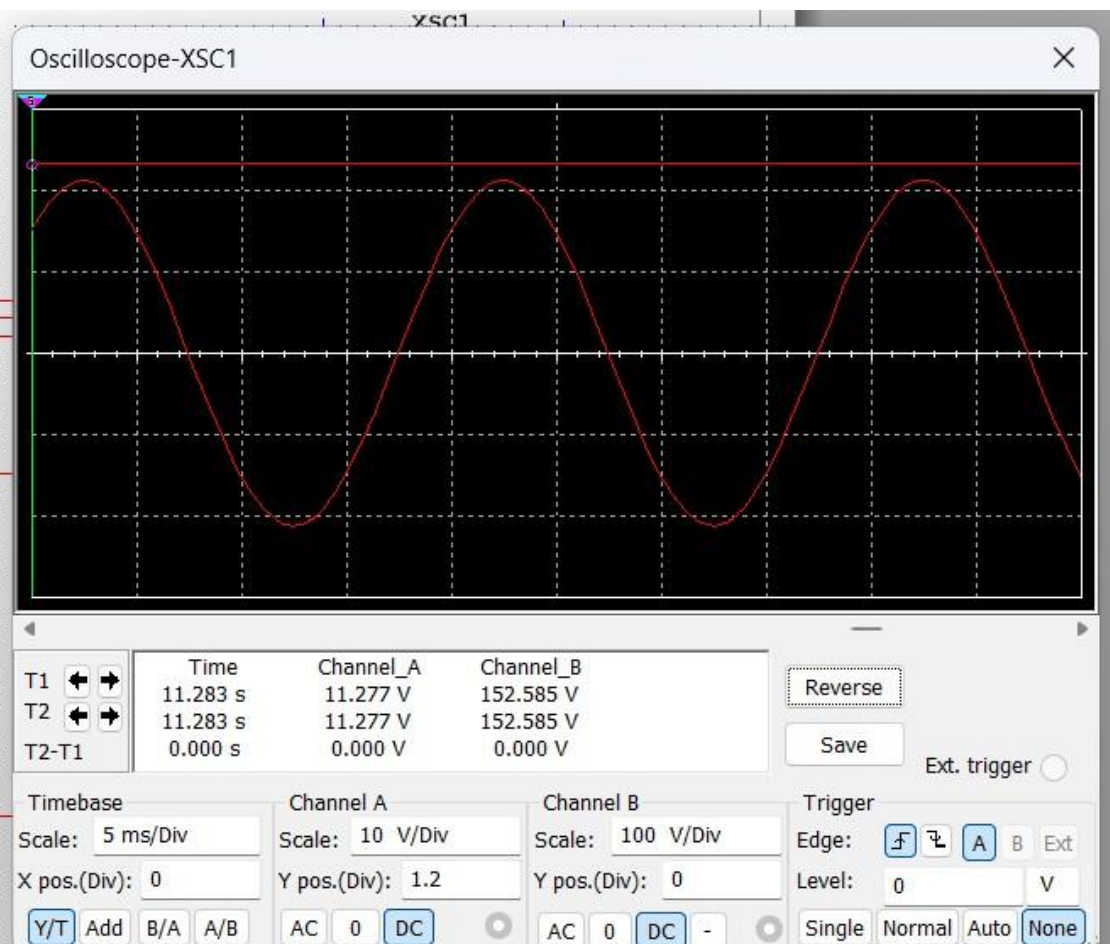
putting the variable resistance R2=5000



putting the variable resistance R2=20



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observation

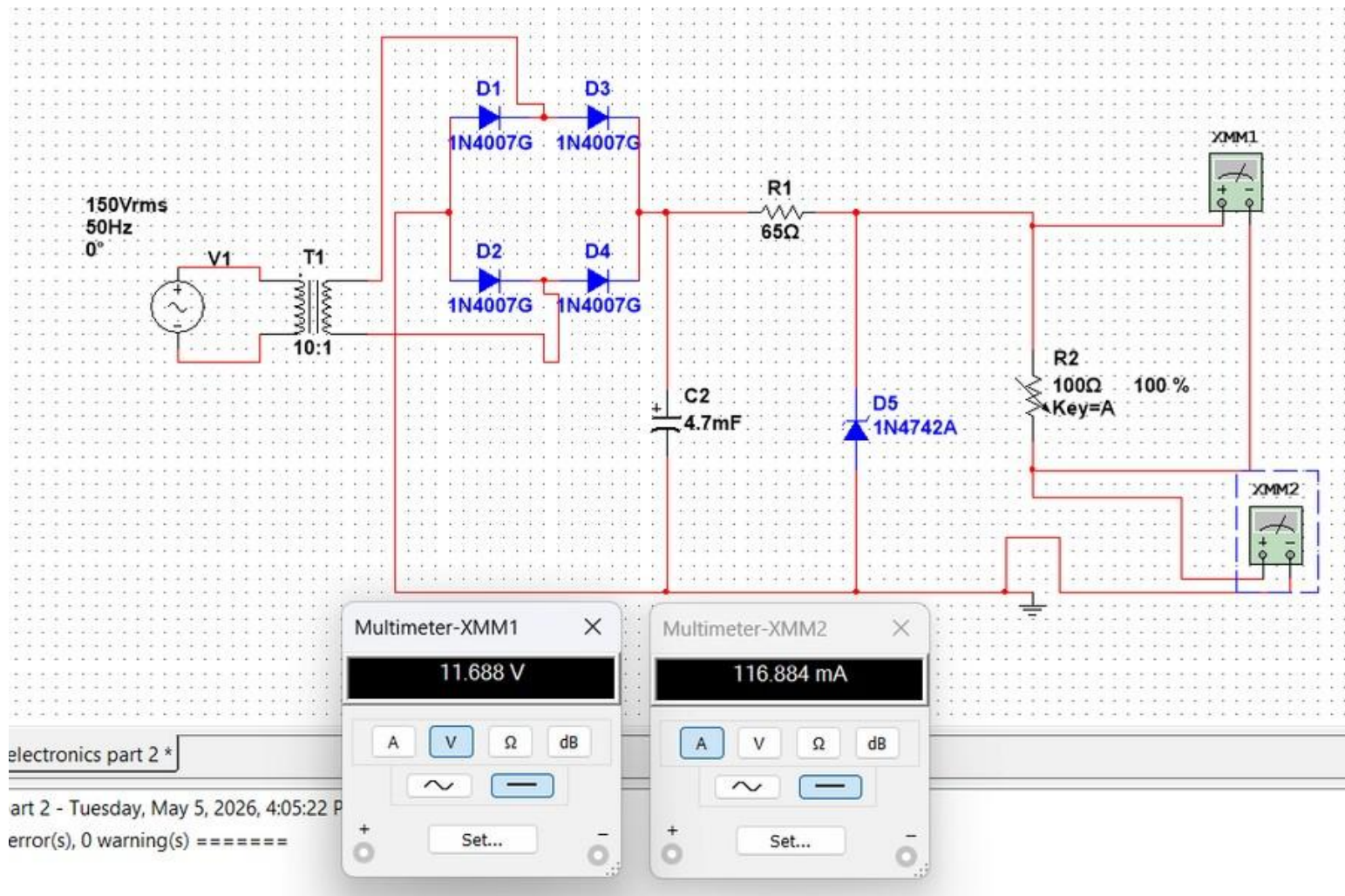
Despite the load resistance changing (from 20Ω to 5000Ω), and the current changing (from 564mA to 2.289mA), the output voltage only changed by:

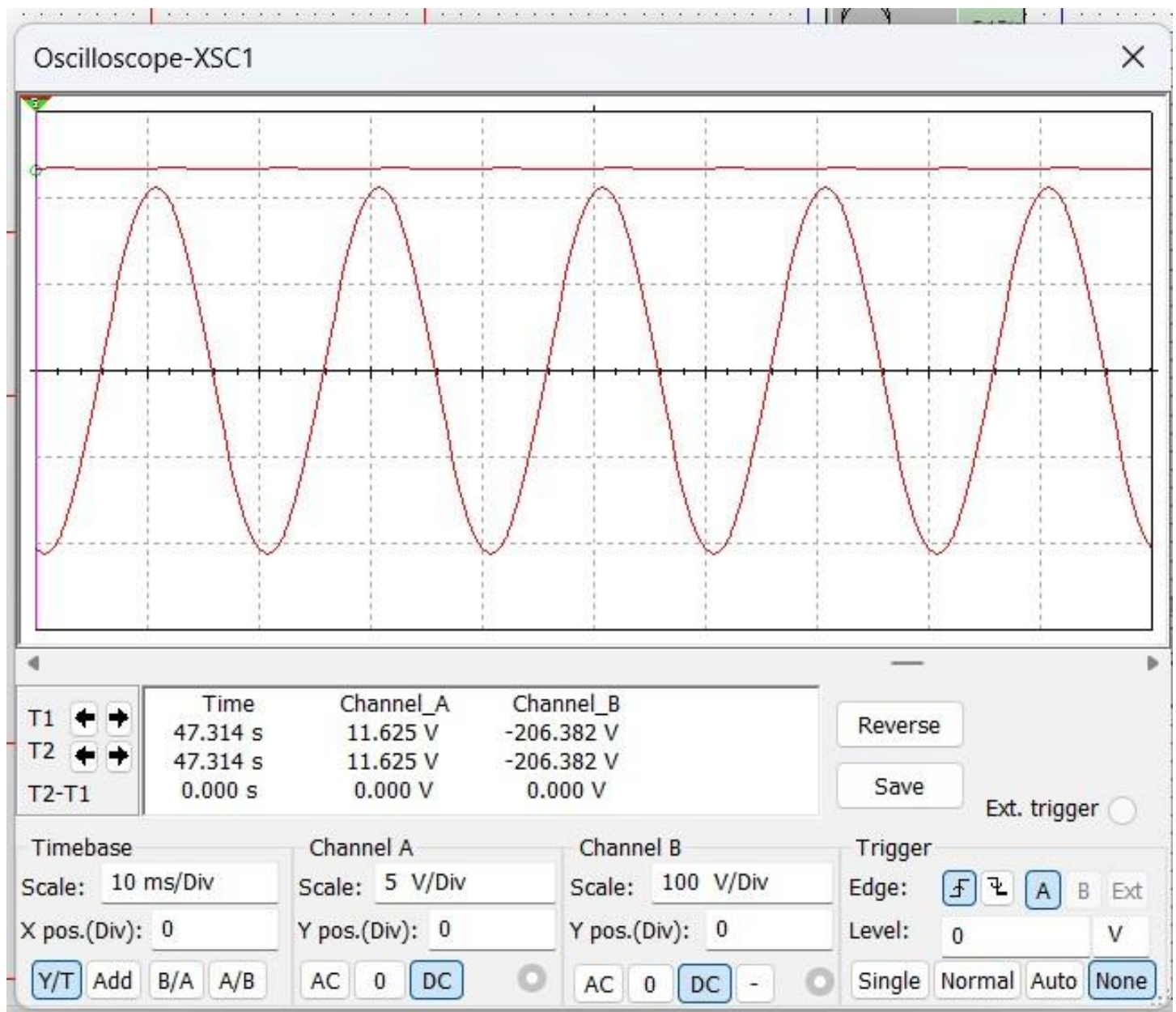
$$\Delta V = 11.435 - 11.28 = 0.155 \text{ V}$$

That is a variation of only 1.37% — this demonstrates excellent voltage regulation by the transistor circuit. The transistor successfully isolates the output from load changes, maintaining a nearly constant voltage across a very wide range of load conditions

removing the transistor and c1

after removing the transistor now R1 must supply ALL the load current directly. With $R_{load} = 100\Omega$ needing about 120mA, R1 must be small enough to allow that current to flow thats why we put it by value 65ohm



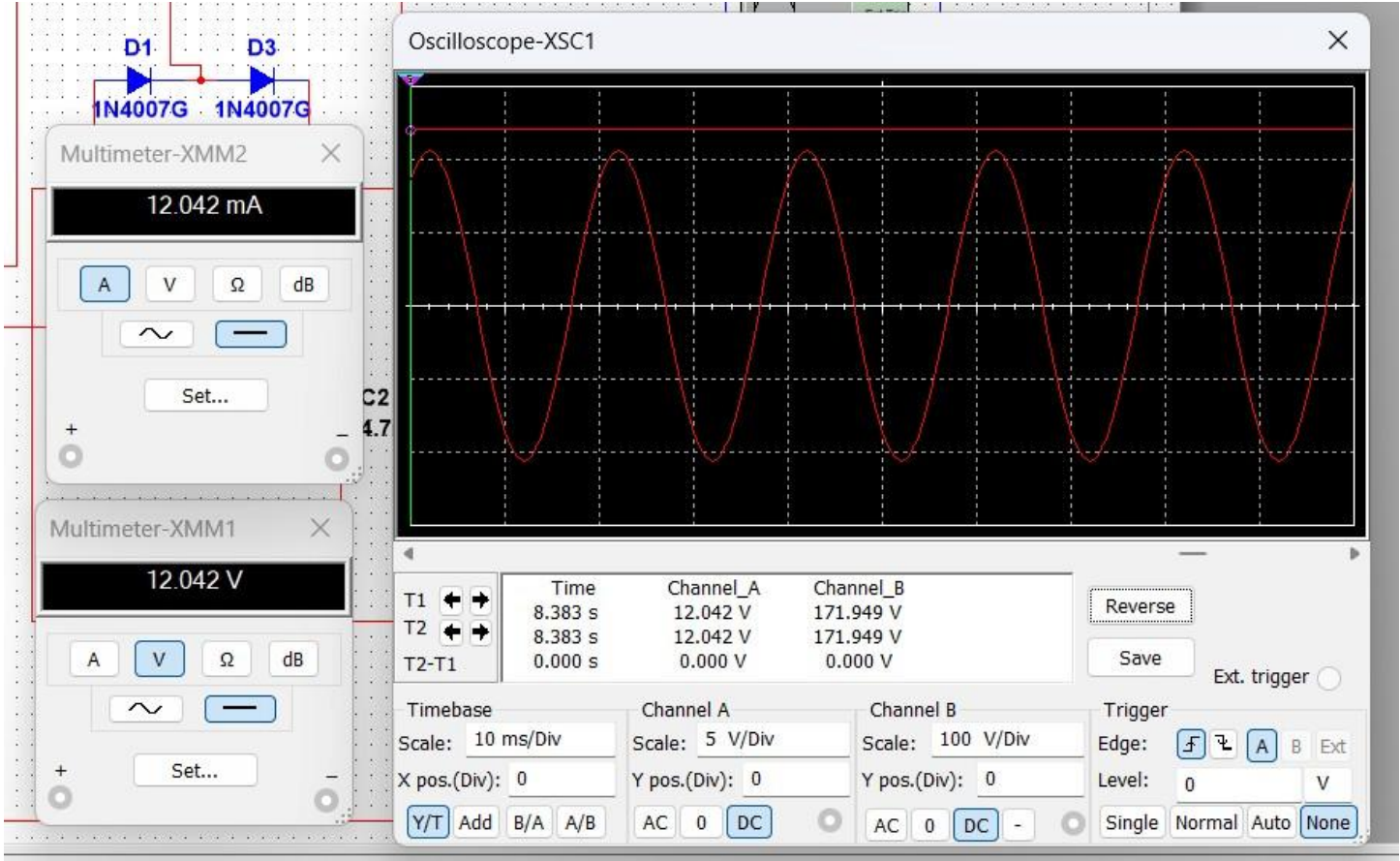
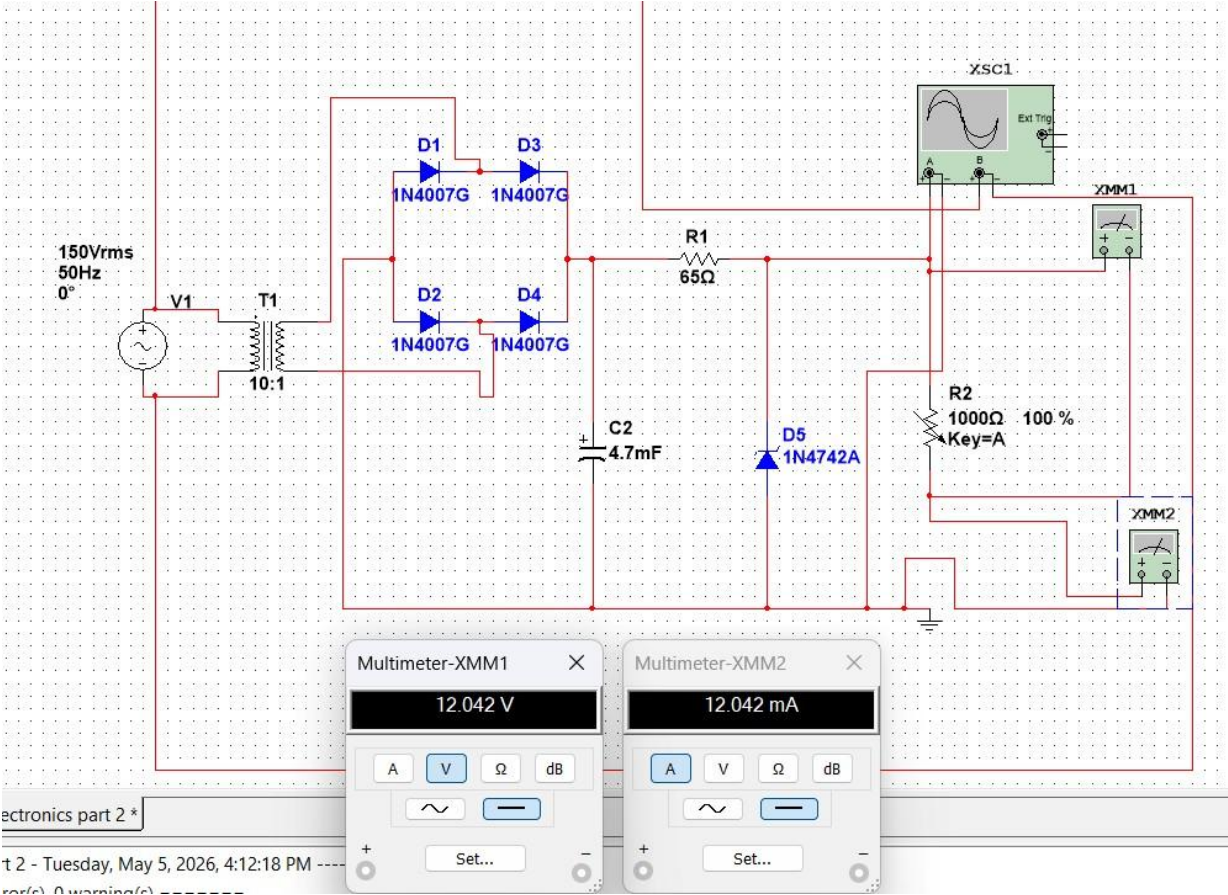


Why is the Output Not Exactly 12V?

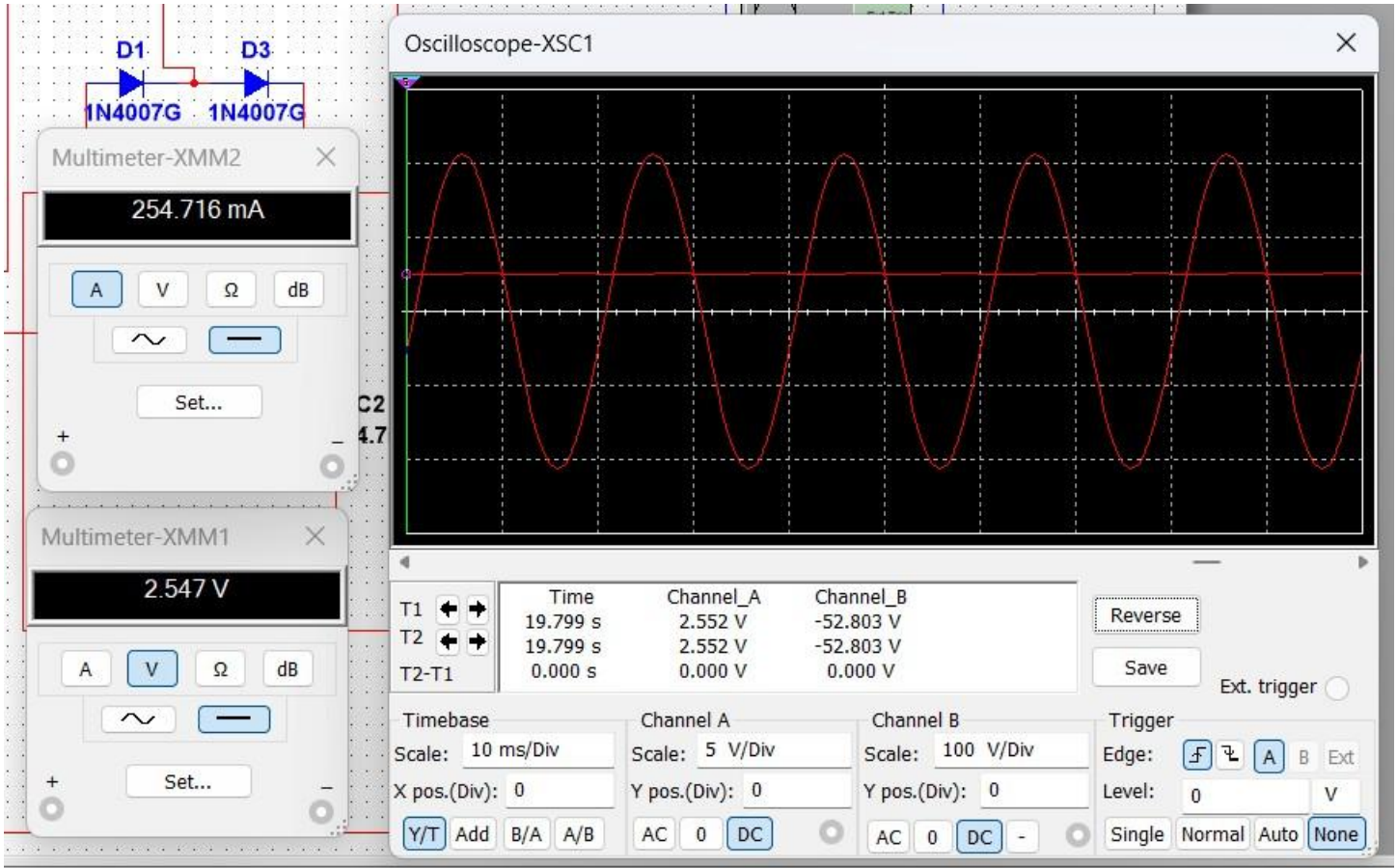
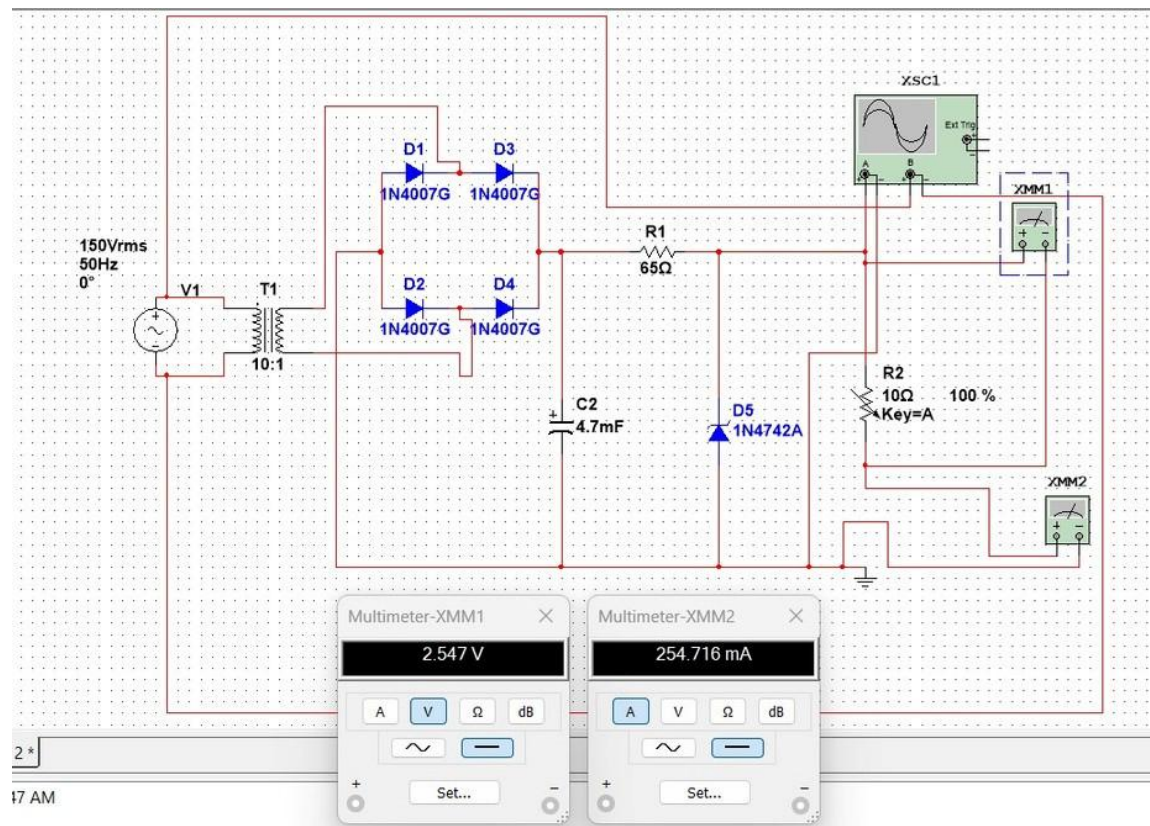
A real Zener diode has a small dynamic resistance (R_z). This means its voltage is not perfectly fixed at 12V so it shifts slightly depending on the current flowing through it

$$V_z = V_z' + R_z \times I_z$$

putting the variable resistance R2=1000



putting the variable resistance R2=10



observation

at 1000 ohm resistance : When the load resistance is very high, the load demands very little current nearly 12mA and most of the current from R1 flows through the Zener diode, keeping it well inside its breakdown region. The Zener successfully regulates and the output is 12.042V , slightly above 12V because the high Zener current raises its voltage slightly due to the Zener internal resistance.

at 10 ohm resistance: the circuit completely fails The 10Ω load demands a very large current ($I = V_z/R$) = **(12/10) =1200mA** But R1 = 65Ω can only supply a maximum of:

$$(19.8-12)/65 = 120\text{mA}$$

The Zener completely exits its breakdown region and stops regulating entirely. The output voltage collapses to 2.547V

Q:8

No. Connecting the load directly across the Zener is not better than using the transistor

At first , the Zener only circuit seems acceptable —at resistance (1000Ω) it actually gives a closer reading to 12V than the transistor circuit, because there is no V_{BE} drop. this advantage disappears completely under(10Ω) resistance.

in the Zener-only circuit, R1 must supply all the load current directly At $R_{load} = 10\Omega$, the load demands 1200mA but $R1 = 65\Omega$ can only deliver 120mA maximum. The load steals all the current from R1, the Zener receives zero current, exits its breakdown region completely, and the output collapses to **2.547V**

With the transistor, the situation is completely different. R1 only supplies a small bias current (16.6mA) to the Zener and the transistor base. The transistor's current gain (β) then amplifies this into a large load current, maintaining **11.28V** even at very low load resistances.

Q:9

the Zener-only circuit is not just inferior in performance, it is also potentially dangerous to the Zener diode itself, The 1N4742A Zener has a maximum power rating of **1 Watt**, meaning the maximum safe current is **83mA** . The danger occurs at light load or no load in the Zener-only circuit. With $R1 = 65\Omega$, when the load resistance is very high, almost all current from $R1$ flows through the Zener and current will be around **120mA** so the power in the zener will be **1.44 Watt**. This exceeds the Zener's 1W maximum power and would burn it out in real hardware. $R1$ must be small to supply enough current to the load at heavy loads , but this burns the Zener at light loads also $R1$ must be large to protect the Zener at light loads , but this causes output voltage collapse at heavy loads **There is no single value of $R1$ that satisfies both conditions safely**

so final answer is The Zener-only design fails under heavy loads and risks destroying the Zener under light loads. The transistor is essential